

Discrete Structures

Assignment: Set Theory

Total Marks: 40

Release Date: **15th September**

Deadline: **19th September**, 11:59 p.m.

Instructions

- The marks for each question (or sub-question) have been specified alongside it.
- Take care to ensure that if you use variables that have not been defined in the question, you state them clearly, and state the domain they belong to clearly as well (if applicable).
- Answer all questions precisely, maintaining mathematical rigor as much as possible.
- For proof-based questions, all necessary steps and justifications must be provided. A final answer without sufficient reasoning may not receive full credit.
- When proving equality of two sets, use a mathematically rigorous argument, such as proving both inclusions or using an equivalent valid method.
- Venn diagrams may be used to illustrate set relationships, but a diagram alone is not considered a proof unless explicitly stated otherwise.
- For questions involving functions or bijections, clearly specify the domain and codomain and verify the required properties.
- Both digital (Written, Not Typed) and handwritten submissions will be accepted.
- Submissions are to be made on Moodle with the format:

$\langle \text{rollnumber} \rangle$ DS A#

(e.g., 2023113019 DS A2).

- Plagiarism and unauthorized use of external or generated solutions shall be penalized heavily. Solutions should be written in your own words and should reflect your own reasoning.

Guide: The Element Method

Proving Set Inclusions and Equalities

What is the Element Method?

The element method is a direct way of proving statements about sets. The main idea is to work with an arbitrary element of a set and use the definitions of set operations to determine where that element must belong.

To prove an inclusion

$$X \subseteq Y,$$

take an arbitrary element $x \in X$, and show that $x \in Y$. Since x was chosen arbitrarily, this proves that every element of X is also an element of Y .

To prove a set equality

$$X = Y,$$

prove both inclusions:

$$X \subseteq Y \quad \text{and} \quad Y \subseteq X.$$

Useful Membership Definitions

When using the element method, translate set operations into statements about membership. For example,

$$\begin{aligned} x \in A \cup B &\iff (x \in A) \vee (x \in B), \\ x \in A \cap B &\iff (x \in A) \wedge (x \in B), \\ x \in A - B &\iff (x \in A) \wedge (x \notin B). \end{aligned}$$

These membership conditions can then be used step by step to prove the required inclusion.

Worked Example

Prove, using the element method, that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Solution.

We prove the two inclusions separately.

1. Prove that

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C).$$

Let x be an arbitrary element of $A \cup (B \cap C)$.

Then

$$x \in A \cup (B \cap C).$$

By the definition of union,

$$x \in A \quad \text{or} \quad x \in B \cap C.$$

If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$. Hence

$$x \in (A \cup B) \cap (A \cup C).$$

If $x \in B \cap C$, then $x \in B$ and $x \in C$. Therefore,

$$x \in A \cup B \quad \text{and} \quad x \in A \cup C,$$

so again

$$x \in (A \cup B) \cap (A \cup C).$$

Thus,

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C).$$

2. Prove that

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C).$$

Let x be an arbitrary element of

$$(A \cup B) \cap (A \cup C).$$

Then

$$x \in A \cup B \quad \text{and} \quad x \in A \cup C.$$

Suppose first that $x \in A$. Then immediately

$$x \in A \cup (B \cap C).$$

Otherwise, $x \notin A$. Since $x \in A \cup B$, we must have

$$x \in B.$$

Similarly, since $x \in A \cup C$, we must have

$$x \in C.$$

Therefore,

$$x \in B \cap C,$$

and hence

$$x \in A \cup (B \cap C).$$

Thus,

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C).$$

Since both inclusions have been proved,

$$\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}.$$

A Useful Checklist

When asked to prove a set equality using the element method:

1. Start with the first inclusion.
2. Let x be an arbitrary element of the set on the left.
3. Write what $x \in$ that set means using the definitions of the set operations.
4. Reason from these membership conditions until you obtain $x \in$ the set on the right.
5. Repeat the process in the opposite direction.
6. Conclude equality only after both inclusions have been established.

Important: Do not choose a particular value for x . The element x should be arbitrary. The goal is to show that the argument works for every possible element of the set.

Problem 1: (2 + 2 + 2 + 2 = 8)

A university runs a Mathematics workshop (M) and a Computing workshop (T) for a group of 80 students, labeled $U = \{1, 2, \dots, 80\}$. Suppose

$$|M| = 46, \quad |T| = 35, \quad |M \cap T| = 18.$$

(a) **[2 marks]**

Determine

$$|M \cup T|, \quad |M - T|, \quad |T - M|.$$

(b) **[2 marks]**

Let E denote the set of students who attend exactly one of the two workshops. Express E in terms of M and T using set operations, and determine $|E|$.

Now, among the students in E , suppose $C \subseteq E$ is the subset of students who also attend a third, optional Career-Skills seminar, with

$$|C| = 20, \quad |C \cap M| = 9.$$

(c) **[2 marks]**

Determine $|C - M|$ and $|C \cap T|$ (using $C \subseteq E$, so $C \cap M$ and $C \cap T$ partition C).

(d) **[2 marks]**

Determine whether each of the following statements is true or false. Give a brief justification for each answer:

$$C \subseteq E, \quad C \in \mathcal{P}(E).$$

Problem 2: (4 + 3 = 7)

Let A , B , and C be arbitrary subsets of a universal set U .

(a) **[4 marks]**

Prove, using the element method, that

$$A - (B \cup C) = (A - B) \cap (A - C).$$

Your proof must establish both inclusions:

$$A - (B \cup C) \subseteq (A - B) \cap (A - C)$$

and

$$(A - B) \cap (A - C) \subseteq A - (B \cup C).$$

(b) [3 marks]

Prove that

$$A \subseteq B \iff A \cup B = B.$$

Problem 3: (2 + 3 + 2 = 7)

(a) [2 marks]

Let

$$X = \{a, b, c\}.$$

List $\mathcal{P}(X)$ explicitly, and determine

$$|\mathcal{P}(\mathcal{P}(X))|$$

without listing all of its elements.

(b) [3 marks]

Prove that for every finite set S with $|S| = n$, we have

$$|\mathcal{P}(S)| = 2^n.$$

Your proof should explain the independent “in” or “out” choice for each element of S .

(c) [2 marks]

In the ZFC cumulative hierarchy, each new stage is built only as the power set of the stage before it (and, at limit stages, as a union of all earlier stages). In two or three sentences, explain why this construction guarantees that no set can ever contain itself.

Problem 4: (3 + 2 + 3 = 8)

(a) [3 marks]

Determine whether each function is injective, surjective, both, or neither. Justify each answer briefly:

$$f_1 : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f_1(x) = x^2,$$

and

$$f_2 : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}, \quad f_2(x) = x^2.$$

(b) [2 marks]

Give an explicit function

$$\mathbb{N} \rightarrow \mathbb{N}$$

that is injective but not surjective, and an explicit function

$$\mathbb{N} \rightarrow \mathbb{N}$$

that is surjective but not injective. Justify each briefly.

(c) [3 marks]

Let

$$h : \mathbb{R} \rightarrow \mathbb{R}$$

be given by

$$h(x) = 2x + 3.$$

Prove that h is bijective by verifying injectivity and surjectivity directly from the definitions.

Problem 5: (3 + 2 = 5)

Let

$$E = \{n \in \mathbb{Z} : 4 \mid n\},$$

that is, E is the set of all integers divisible by 4.

(a) [3 marks]

Prove that E is countably infinite by constructing an explicit bijection

$$f : \mathbb{Z} \rightarrow E.$$

Clearly define f and verify both injectivity and surjectivity.

(b) [2 marks]

Prove that

$$E \times \{0, 1\}$$

is countably infinite. You may either construct an explicit bijection with $\mathbb{N}$ or provide a suitable enumeration and justify why every element occurs exactly once.

Problem 6: (2 + 1 + 2 = 5)

Let

$$A = \mathbb{R} \setminus \{1\}.$$

Define

$$f : A \rightarrow A, \quad f(x) = \frac{x+1}{x-1}.$$

Let

$$g : \mathbb{R} \rightarrow \mathbb{R}$$

be defined by

$$g(x) = x^2.$$

(a)

[2 marks]

Compute the composition

$$(f \circ f)(x)$$

and simplify your answer.

(b)

[1 mark]

Using your result from part (a), determine the inverse function f^{-1} in terms of f .

(c)

[2 marks]

Find the composition

$$(g \circ f)(x).$$

Is $g \circ f$ an injective function? Justify your answer.

— *End of Assignment* —